Adaptive Fabric Surface Defect Detection and Localization using Classical Texture Descriptors

Weekly Progress Report — Week 5

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1 Introduction

Week 5 marks the transition from the Fixed Pipeline baseline to our proposed solution: the **Weave-Adaptive Pipeline** (Phase 2). Following our Week 4 evaluation demonstrating that rigid 15×15 inspection footprints diluted localized defect signals on fine weaves, our focus shifted toward developing a decoupled multi-scale detection architecture. This system deterministically derives filter parameters from each fabric's fundamental weave repeat pitch and employs unsupervised statistical gating to prevent false alarm proliferation.

2 Weave Pitch Estimation via 2D Autocorrelation

To replace arbitrary filter sizing with a deterministic approach, we implemented an automated weave pitch estimation module (`src/adaptive/scale_estimation.py`).

2.1 Pitch Estimation via 2D FFT

Periodic thread structures produce prominent harmonic peaks in the spatial autocorrelation domain. Using the Wiener–Khinchin theorem, 2D autocorrelation surfaces are computed from defect-free reference scans via 2D FFT. Extracting 1D radial profiles through the central peak identifies the first secondary peak, whose spatial distance corresponds directly to the physical thread repeat period (weave pitch P, in pixels).

2.2 Derived Inspection Scales

The measured pitch deterministically dictates inspection kernel dimensions across the 7 AITEX weaves:

- **Fabrics 00, 01, and 04 (Plain and Twills):** Fundamental pitch of 15 px, mapped to a 15×15 footprint.
- **Fabrics 02 and 06 (Coarse Twill and Satin):** Repeat pitch of 13 px, mapped to a 13×13 footprint.
- **Fabric 05 (Warp-Faced Stripe Twill):** Repeat pitch of 11 px, mapped to an 11×11 footprint.
- **Fabric 03 (Fine Herringbone):** Tight pitch of 9 px, mapped to a compact 9×9 footprint.

This dynamic matching ensures filter kernels cancel nominal background weave rhythms without blurring localized yarn flaws.

3 Three-Branch Multi-Scale Detection Architecture

Because textile defects span micro-knots (1–10 px) to macro tears (hundreds of pixels), no single filter can capture all flaw typologies. We engineered a parallel three-branch architecture:

3.1 Branch Specialization

- **Micro Branch (Top-Hat Filtering):** Evaluates White and Black Top-Hat operators at native weave pitch ($1T$) and sub-pitch ($0.5T$) to suppress nominal weave patterns and isolate localized yarn disruptions (knots, slubs, pinholes).
- **Macro Branch (Gaussian Spatial Envelope):** Employs a broad 2D Gaussian filter ($W = 201, \sigma = 50$) to estimate the smooth illumination envelope, exposing wide tears, voids, and horizontal shade bars.
- **Warp Branch (Column Profiling):** Integrates pixel intensities vertically along thread columns (1×35 filter), highlighting continuous warp misdraws while suppressing static camera CCD artifacts.

3.2 Statistical Threshold Selection and Fusion

Evaluating textbook Otsu thresholding revealed a severe failure: because sparse defects occupy $< 0.5\%$ of image pixels, Otsu's variance maximization partitioned normal fabric noise, causing an unacceptable 71.02% false positive rate. To resolve this, we formulated an unsupervised percentile gate ($P_{99.99}$) calibrated strictly on defect-free training rolls, bounding clean noise below 0.01%. Candidate detections across all three branches are then merged via pixel-wise Boolean OR fusion.

4 Work Completed This Week

- Implemented 2D FFT spatial autocorrelation to deterministically measure weave repeat pitch across all 7 fabric types.
- Formulated and integrated the three parallel detection branches (Micro Top-Hat, Macro Gaussian envelope, and Warp column profiling).
- Diagnosed the theoretical failure of Otsu thresholding on sparse defects and implemented the unsupervised $P_{99.99}$ clean-roll calibration gate.
- Implemented base multi-branch Boolean OR fusion to unify candidate anomaly maps.

5 Plan for Next Week

With the adaptive detection branches established, next week will focus on downstream post-processing and benchmark validation:

- Develop intelligent morphological cleanup (geometry-gated tear closing, small dust speck elimination, sensor blemish removal).
- Implement 8-connectivity connected component localization for bounding-box extraction and severity scoring.
- Execute full 81-roll comparative benchmarking against the Fixed Control baseline.

End of Week 5 Report